

Cheung Ka Fai, Cyrus

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EXPERIENCE

Castco Testing Centre Limited

Sep 2023 – Present

AI Programmer

- Pioneered phone customer support using Twilio and OpenAI. Developed an LLM agent that enabled dynamic conversations, answered questions using company knowledge, and routed calls to appropriate departments.
- Led the development of a construction equipment detection model fine-tuned from YOLO, using a labeled internal video dataset. Improved video stream tracking performance in complex environments using the SORT algorithm.
- Increased document handling efficiency by 50% using Google Cloud Platform AI OCR, with over 100 labeled documents for custom format recognition.
- Wrote clear and concise documentation of training and testing procedures allowing future teams access to quick and easy guidance on reproducing current results.

PONS.ai

May 2022 – Aug 2022

AI Engineer Intern

- Conducted research on state-of-the-art generative AI models and techniques for image style transfer
- Optimized the AI pipeline by incorporating traditional algorithms and neural networks, resulting in a 2/3 reduction in inference time
- Contributed to the complete development cycle, including data collection, image preprocessing, and experimentation with StyleGAN

EDUCATION

The University of Hong Kong

Sep 2019 – Aug 2023

Bachelor of Arts and Science (Applied AI)

GPA 3.38

- Studied advanced topics in mathematical foundations, machine learning architectures and research methodologies
- Relevant coursework includes Applied Deep Learning, Image Processing and Computer Vision, and Natural Language Processing
- Proposed and researched new Adam-like optimization strategies to improve generalization ability by experimenting with second-moment exponents.

PROJECTS

Context-Aware Machine Translation | *Language Processing, Research, Electron*

- Integrated a local LLM for context-sensitive language translation.
- Enhanced translation quality using state-of-the-art self-refinement methods for linguistic errors and few-shot learning.
- Enabled automatic quality estimation using textual embeddings for manual assessment.
- Productized as standalone desktop software using Electron.

Real-Time Segmentation on Unified Dash Cam Images | *Computer Vision, Pytorch*

- Applied state-of-the-art real-time image segmentation models for parsing road objects.
- Unified multiple street scene datasets, including Cityscapes, BDD100K, and Mapillary Vistas, to cover diverse environments.
- Achieved over 60 frames per second on consumer devices with high accuracy.

Fine-Tuned Qwen for Code Change Summarization | *Huggingface, Fine-tuning*

- Fine-tuned Qwen2.5 Coder to generate Git commit messages based on code changes.
- Used a dataset of 300,000 high-quality commits for training, resulting in improved instruction-following.
- Tracked experiment hyperparameters using Weights & Biases.

TECHNICAL SKILLS

Machine Learning: Object Detection, Semantic Segmentation, Retrieval-Augmented Generation (RAG)

Tools: Huggingface, PyTorch, Amazon Web Services (AWS), Llama.cpp